CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

LUIS CORREIA DAIRY

11132 SCHULTZ RD

EL NIDO CA 95317

NOTES:

PREPARED BY:

CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Luis Correia Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
11132 Schultz RD	El Nido	Merced	95317

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X075-X073-X003-XXXX

B. OPERATORS

Correia, Luis

Operator name: Correia, Luis

Telephone no.: (209) 201-1520 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11132 Schultz RD	El Nido	CA	95344

This operator is responsible for paying permit fees.

C. OWNERS

Cabral, Bros.

Legal owner name: Cabral, Bros.

Telephone no.: (209) 364-2603 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
1803 Alderwood CT	San Mateo	CA	94402

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	205	42	190	60	55	0
Number under roof	0	0	0	0	0	0
Maximum number	205	42	190	60	55	0
Average number	205	42	190	60	55	0
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 8,654.39 *tons per reporting period*

Total nitrogen from manure: 102,324.29 *lbs per reporting period*

After ammonia losses (30% loss applied): 71,627.00 *lbs per reporting period*

Total phosphorus from manure: 16,749.30 *lbs per reporting period*

Total potassium from manure: 38,990.02 *lbs per reporting period*

Total salt from manure: 106,182.15 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,900,000 *gallons*

Total nitrogen generated: 29,031.59 *lbs*

Total phosphorus generated: 8,390.73 *lbs*

Total potassium generated: 61,188.04 *lbs*

Total salt generated: 261,966.24 *lbs*

	12,900,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	12,900,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	33	33	1	none	X075-X073-X003-XXXX
Field 2	112	112	2	both	X075-X073-X004-XXXX
Totals for areas that were used for application	112	112	2		
Totals for areas that were not used for application	33	33	1		
Land application area totals	145	145	3		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

01/01/2009: Almond, in shell

Crop: Almond, in shell Acres planted: 33 Plant date: 01/01/2009

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/30/2023	45.00 *ton*	Dry-weight		10.0	35,900.00	3,800.00	30,600.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.50	195.00	30.00	150.00	0.00
Total actual harvest content	1.36	88.12	9.33	75.11	257.73

Field 2

Field name: Field 2

Field 2

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 112 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	1,580.00 *ton*	Dry-weight		55.9	17,700.00	2,800.00	17,400.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.11	220.23	34.84	216.50	1,082.50

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 112 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	2,325.00 *ton*	Dry-weight		65.8	10,300.00	2,700.00	13,800.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.76	146.25	38.34	195.95	837.75

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 01/01/2009: Almond, in shell

Field name: Field 1

Crop: Almond, in shell

Plant date: 01/01/2009

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.93	0.00	0.00	679.95	4,620,000.00 *gal*
Application event totals		9.93	0.00	0.00	679.95	

03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.65	0.00	0.00	592.53	4,026,000.00 *gal*
Application event totals		8.65	0.00	0.00	592.53	

04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.51	0.00	0.00	582.81	3,960,000.00 *gal*
Application event totals		8.51	0.00	0.00	582.81	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.37	0.00	0.00	573.10	3,894,000.00 *gal*
Application event totals		8.37	0.00	0.00	573.10	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.94	0.00	0.00	611.96	4,158,000.00 *gal*
Application event totals		8.94	0.00	0.00	611.96	

Field 1 - 01/01/2009: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.09	0.00	0.00	553.67	3,762,000.00 *gal*
Application event totals		8.09	0.00	0.00	553.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.22	0.00	0.00	631.38	4,290,000.00 *gal*
Application event totals		9.22	0.00	0.00	631.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.22	0.00	0.00	631.38	4,290,000.00 *gal*
Application event totals		9.22	0.00	0.00	631.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.08	0.00	0.00	621.67	4,224,000.00 *gal*
Application event totals		9.08	0.00	0.00	621.67	

Field 2 - 10/20/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	165.87	79.90	23.77	0.00	728.00 *ton*
Application event totals		165.87	79.90	23.77	0.00	

Field 2 - 10/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	54.94	12.41	90.98	487.29	3,000,000.00 *gal*
Irrigation Well	Ground water	7.22	0.00	0.00	494.35	11,400,000.00 *gal*
Application event totals		62.16	12.41	90.98	981.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	47.57	12.54	86.73	473.88	3,000,000.00 *gal*
Irrigation Well	Ground water	6.80	0.00	0.00	465.73	10,740,000.00 *gal*
Application event totals		54.37	12.54	86.73	939.61	

Field 2 - 06/08/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.46	0.00	0.00	40.95	19,627,979.00 *gal*
Application event totals		1.46	0.00	0.00	40.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.84	0.00	0.00	468.33	10,800,000.00 *gal*
Application event totals		6.84	0.00	0.00	468.33	

Field 2 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.94	23.90	176.30	658.96	3,300,000.00 *gal*
Irrigation Well	Ground water	7.39	0.00	0.00	505.80	11,664,000.00 *gal*
Canal	Surface water	1.13	0.00	0.00	31.57	15,132,189.00 *gal*
Application event totals		83.46	23.90	176.30	1,196.33	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.70	0.00	0.00	458.97	10,584,000.00 *gal*
Canal	Surface water	0.44	0.00	0.00	12.36	5,925,188.00 *gal*
Application event totals		7.14	0.00	0.00	471.33	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	5.62	0.00	0.00	385.07	8,880,000.00 *gal*
Canal	Surface water	0.66	0.00	0.00	18.36	8,802,233.00 *gal*
Application event totals		6.28	0.00	0.00	403.44	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.76	26.07	192.32	718.86	3,600,000.00 *gal*
Irrigation Well	Ground water	6.48	0.00	0.00	443.36	10,224,000.00 *gal*
Canal	Surface water	0.66	0.00	0.00	18.39	8,814,291.00 *gal*
Application event totals		88.89	26.07	192.32	1,180.61	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.59	0.00	0.00	16.44	7,877,981.00 *gal*
Application event totals		0.59	0.00	0.00	16.44	

Field 2 - 06/08/2023: Corn, silage

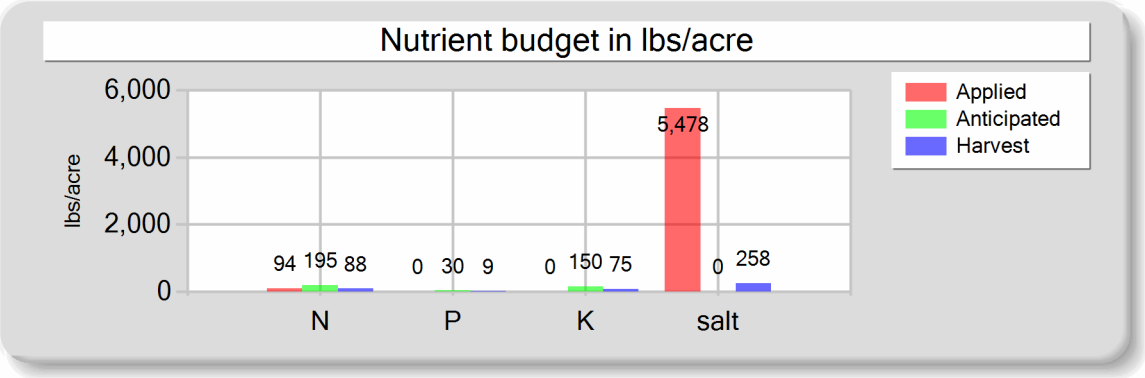
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.54	0.00	0.00	15.04	7,208,582.00 *gal*
Application event totals		0.54	0.00	0.00	15.04	

B. NUTRIENT BUDGET

Field 1 - 01/01/2009: Almond, in shell

Field name: Field 1 Crop: Almond, in shell Plant date: 01/01/2009



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	80.01	0.00	0.00	5,478.46
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	94.01	0.00	0.00	5,478.46
Anticipated crop nutrient removal	195.00	30.00	150.00	0.00
Actual crop nutrient removal	88.12	9.33	75.11	257.73
Nutrient balance	5.89	-9.33	-75.11	5,220.73
Applied to removed ratio	1.07	0.00	0.00	21.26

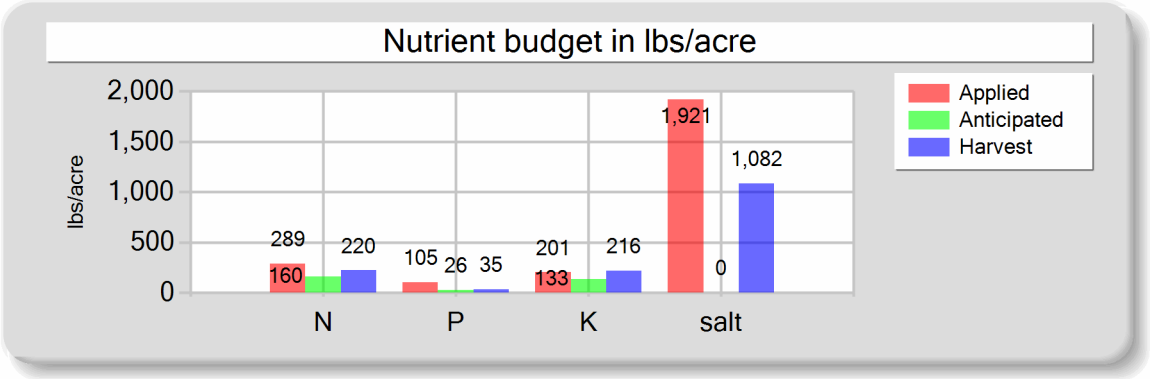
Fresh water applied	
37,224,000.00	*gallons*
1,370.83	*acre-inches*
41.54	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 10/20/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	165.87	79.90	23.77	0.00
Process wastewater	102.51	24.95	177.70	961.17
Fresh water	14.02	0.00	0.00	960.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.40	104.85	201.47	1,921.25
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	220.23	34.84	216.50	1,082.50
Nutrient balance	69.17	70.01	-15.03	838.75
Applied to removed ratio	1.31	3.01	0.93	1.77

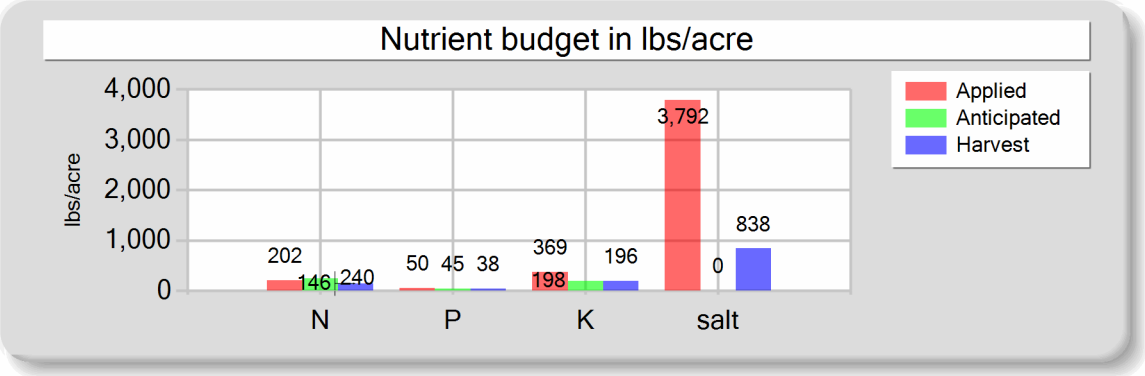
Fresh water applied	
22,140,000.00	*gallons*
815.34	*acre-inches*
7.28	*inches/acre*

Process wastewater applied	
6,000,000.00	*gallons*
220.96	*acre-inches*
1.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/08/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	156.70	49.97	368.62	1,377.82
Fresh water	38.50	0.00	0.00	2,414.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	202.20	49.97	368.62	3,792.46
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	146.25	38.34	195.95	837.75
Nutrient balance	55.95	11.63	172.67	2,954.71
Applied to removed ratio	1.38	1.30	1.88	4.53

Fresh water applied	
125,540,443.00	*gallons*
4,623.23	*acre-inches*
41.28	*inches/acre*

Process wastewater applied	
6,900,000.00	*gallons*
254.10	*acre-inches*
2.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall 2022 Corral Manure

Sample and source description: Fall 2022 Corral Manure

Sample date: 10/10/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 28.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,800.00	5,100.00	23,700.00							
DL	0.01	0.01	0.01							

Scraped Manure

Sample and source description: Scraped Manure

Sample date: 10/04/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 45.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,700.00	3,000.00	6,800.00							
DL	0.01	0.01	0.01							

SSB Manure

Sample and source description: SSB Manure

Sample date: 12/12/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,800.00	15,800.00	4,700.00							
DL	0.01	0.01	0.01							

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 2022

Sample and source description: Lagoon Q4 2022

Sample date: 12/01/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	244.00	167.00	0.00	1.80	55.50	407.00								4,690.00	2,180
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	926.00	190.00	0.00	1.00	168.00	659.00								5,230.00	3,560
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	211.00	118.00	0.00	1.80	56.10	388.00								4,320.00	2,120
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 08/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	304.00	131.00	0.00	0.80	97.20	717.00								5,500.00	2,680
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 12/05/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	218.00	163.00	0.00	0.00	37.00	525.00								5,250.00	2,300
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

C. FRESH WATER ANALYSES

Canal

MID Canal

Sample description: MID Canal

Sample date: 10/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	1.00								32.20	28
DL	0.01	0.01	0.01								0.01	1

Domestic Well

DW Barn

Sample description: DW Barn

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.10	0.00	5.10								658.00	
DL	0.01	0.01	0.01								0.01	

Domestic Well

DW House

Sample description: DW House

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.20	0.00	5.20								634.00	
DL	0.01	0.01	0.01								0.01	

Irrigation Well

IW

Sample description: IW

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.50	0.00	8.50								970.00	
DL	0.01	0.01	0.01								0.01	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 01/01/2009: Almond, in shell

Field 1 - 01/01/2009: Almond, in shell

Field 1 Almonds

Sample and source description: Field 1 Almonds

Sample date: 08/22/2022 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,900.00	3,800.00	30,600.00		0.00
DL	0.01	0.01	0.01		1.00

Field 2 - 10/20/2022: Oats, silage-soft dough

Field 2 Oats A

Sample and source description: Field 2 Oats A

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	2,800.00	16,800.00		8.20
DL	0.01	0.01	0.01		1.00

Field 2 Oats B

Sample and source description: Field 2 Oats B

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,900.00	2,800.00	18,000.00		9.10
DL	0.01	0.01	0.01		1.00

Field 2 - 06/08/2023: Corn, silage

Field 2 - 06/08/2023: Corn, silage

Field 2 Corn

Sample and source description: Field 2 Corn

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,200.00	2,900.00	12,200.00		5.50
DL	0.01	0.01	0.01		1.00

Field 2B Corn

Sample and source description: Field 2B Corn

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,500.00	15,400.00		6.30
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

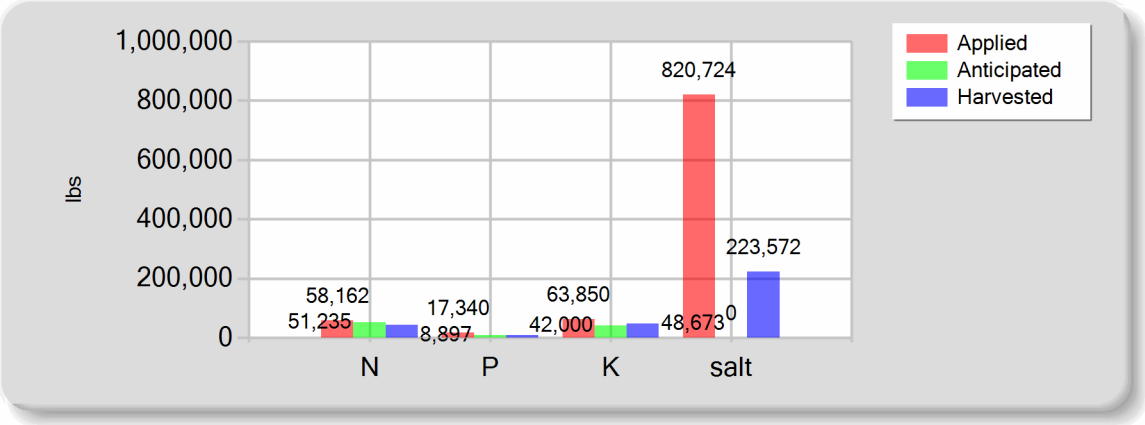
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

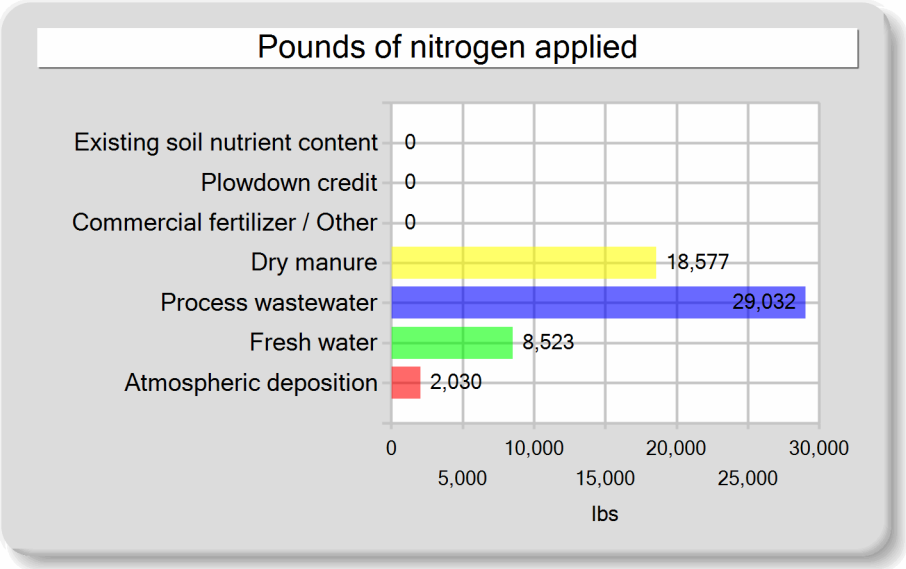
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

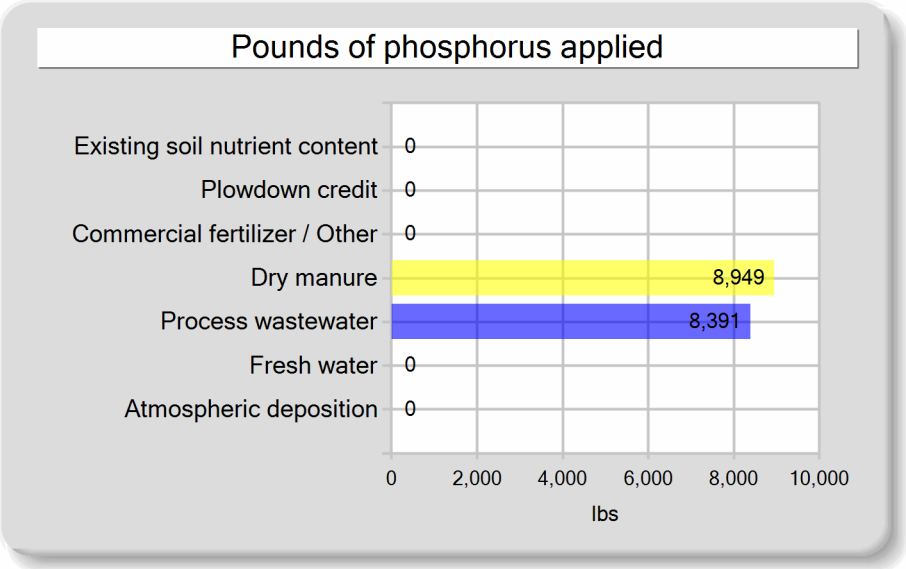
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	18,577.40	8,948.87	2,662.00	0.00
Process wastewater	29,031.59	8,390.73	61,188.04	261,966.24
Fresh water	8,522.54	0.00	0.00	558,757.74
Atmospheric deposition	2,030.00	0.00	0.00	0.00
Total nutrients applied	58,161.52	17,339.60	63,850.05	820,723.98
Anticipated crop nutrient removal	51,235.00	8,897.20	41,999.60	0.00
Actual crop nutrient removal	43,954.00	8,503.58	48,672.68	223,572.42
Nutrient balance	14,207.52	8,836.02	15,177.36	597,151.56
Applied to removed ratio	1.32	2.04	1.31	3.67

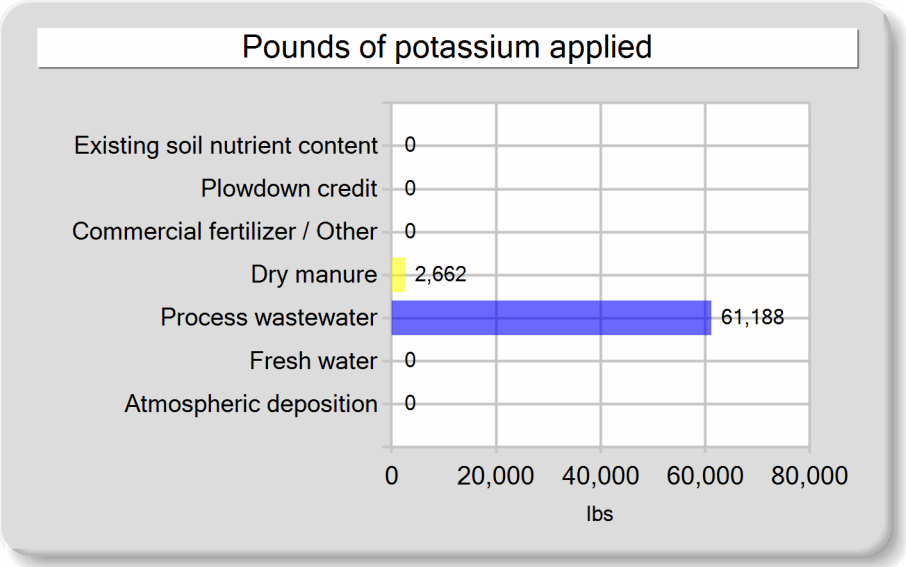
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

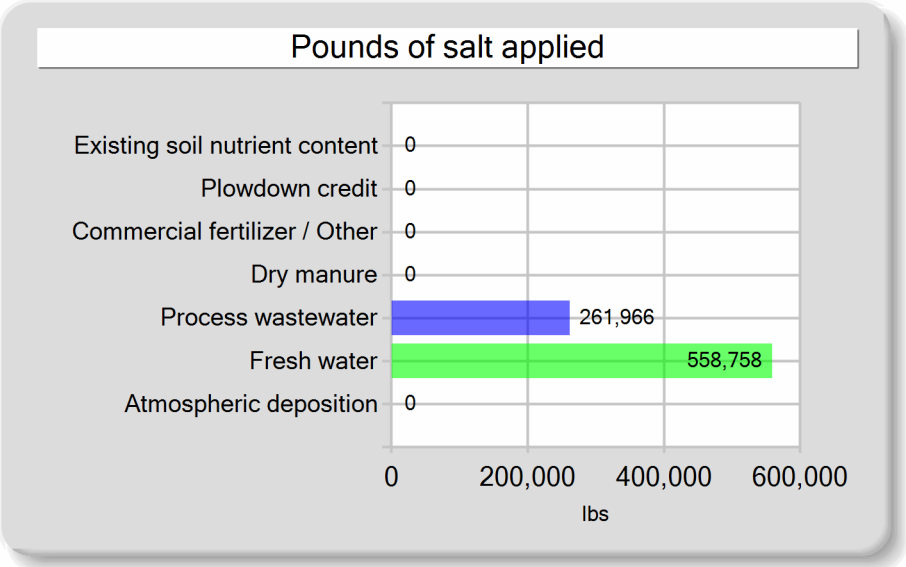


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John Cabral	Louie Correia
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Bros. Cabral	Luis Correia
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 28, 2024	Jun 28, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23L0771
Sampled By: L Correia
Requested By: Luis Correia
Submitted Date: 12/13/23
Reported Date: 12/20/23
Project:
Crop ID:

Luis Correia Dairy
5973 Sugarplum Drive
Atwater CA 95301
00-0021343 08

E-mail:
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW Barn	12/11/23	19:00	658	0.66	5.1	ND		7.6
2	DW House	12/11/23	19:10	634	0.63	5.2	ND		6.9
3	IW	12/11/23	19:25	970	0.97	8.5	ND		7.4

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Luis Correia Dairy
5973 Sugarplum Drive
Atwater CA 95301
00-0021343 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0771
Sampled By: L Correia
Requested By: Luis Correia
Submitted Date: 12/13/23
Reported Date: 12/20/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW Barn	12/11/23	19:00	658	0.66	**5.1**	13.92	1.16
2	DW House	12/11/23	19:10	634	0.63	**5.2**	14.20	1.18
3	IW	12/11/23	19:25	**970**	**0.97**	**8.5**	23.21	1.93

Water for Crop Use	Total	Total	Nitrate
Nutrient Management	Salts	Salts	Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Luis Correia Dairy
5973 Sugarplum Drive
Atwater, CA 95301

Account# 00-0021343
Account Manager: Ben Nydam
Submitted By: Luis Correia

Received: 12/13/2023 7:00
Reported: 12/20/2023 15:13

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0771-01	DW Barn	Ag Water	L Correia		12/11/2023 19:00
23L0771-02	DW House	Ag Water	L Correia		12/11/2023 19:10
23L0771-03	IW	Ag Water	L Correia		12/11/2023 19:25

Default Cooler Temperature on Receipt °C: 5.4
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Luis Correia Dairy
5973 Sugarplum Drive
Atwater, CA 95301

Account# 00-0021343
Account Manager: Ben Nydam
Submitted By: Luis Correia

Received: 12/13/2023 7:00
Reported: 12/20/2023 15:13

Sample Results

Sample: DW Barn
23L0771-01 (Water)

Sampled: 12/11/2023 19:00
Sampled By: L Correia

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.66**	mmhos/cm	0.01	1		12/14/23 15:26	SM 2510 B		BEL0614
Electrical Conductivity umhos	**658**	umhos/cm	10.0	1		12/14/23 15:26	SM 2510 B		BEL0614
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 19:00	Field		BEL0438
Nitrate Nitrogen as NO3N	**5.1**	mg/L	0.1	1	10	12/13/23 18:22	EPA 300.0		BEL0444
Temperature	**25.0**	units	0.0	1		12/14/23 15:26	SM 4500-H+	H	BEL0614
pH	**7.6**	units	1.0	1		12/14/23 15:26	SM 4500-H+	H	BEL0614

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DELLAVALLE™
LABORATORY INC

Luis Correia Dairy
5973 Sugarplum Drive
Atwater, CA 95301

Account# 00-0021343
Account Manager: Ben Nydam
Submitted By: Luis Correia

Received: 12/13/2023 7:00
Reported: 12/20/2023 15:13

Sample Results
(Continued)

Sample: DW House
23L0771-02 (Water)

Sampled: 12/11/2023 19:10
Sampled By: L Correia

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.63**	mmhos/cm	0.01	1		12/14/23 16:03	SM 2510 B		BEL0615
Electrical Conductivity umhos	**634**	umhos/cm	10.0	1		12/14/23 16:03	SM 2510 B		BEL0615
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 19:10	Field		BEL0438
Nitrate Nitrogen as NO3N	**5.2**	mg/L	0.1	1	10	12/13/23 18:42	EPA 300.0		BEL0444
Temperature	**25.0**	units	0.0	1		12/14/23 16:03	SM 4500-H+	H	BEL0615
pH	**6.9**	units	1.0	1		12/14/23 16:03	SM 4500-H+	H	BEL0615

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DELLAVALLE™
LABORATORY INC

Luis Correia Dairy
5973 Sugarplum Drive
Atwater, CA 95301

Account# 00-0021343
Account Manager: Ben Nydam
Submitted By: Luis Correia

Received: 12/13/2023 7:00
Reported: 12/20/2023 15:13

Sample Results
(Continued)

Sample: IW
23L0771-03 (Water)

Sampled: 12/11/2023 19:25
Sampled By: L Correia

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.97**	mmhos/cm	0.01	1		12/14/23 16:05	SM 2510 B		BEL0615
Electrical Conductivity umhos	**970**	umhos/cm	10.0	1		12/14/23 16:05	SM 2510 B		BEL0615
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 19:25	Field		BEL0438
Nitrate Nitrogen as NO3N	**8.5**	mg/L	0.1	1	10	12/13/23 19:02	EPA 300.0		BEL0444
Temperature	**25.0**	units	0.0	1		12/14/23 16:05	SM 4500-H+	H	BEL0615
pH	**7.4**	units	1.0	1		12/14/23 16:05	SM 4500-H+	H	BEL0615

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DELLAVALLE™ LABORATORY INC

Luis Correia Dairy
5973 Sugarplum Drive
Atwater, CA 95301

Account# 00-0021343
Account Manager: Ben Nydam
Submitted By: Luis Correia

Received: 12/13/2023 7:00
Reported: 12/20/2023 15:13

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0444										
Blank (BEL0444-BLK1)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0444-BLK2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0444-BLK3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0444-BLK4)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0444-BS1)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
LCS (BEL0444-BS2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
LCS (BEL0444-BS3)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
Duplicate (BEL0444-DUP1)		**Source: 23L0740-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	0.05		0.1	mg/L		0.05			3.92	10
Duplicate (BEL0444-DUP2)		**Source: 23L0744-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	0.05		0.1	mg/L		0.05			1.98	10
Duplicate (BEL0444-DUP3)		**Source: 23L0700-01**			Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	0.04		0.1	mg/L		0.04			0.00	10
Matrix Spike (BEL0444-MS1)		**Source: 23L0740-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000	0.05	101	90-110		
Matrix Spike (BEL0444-MS2)		**Source: 23L0744-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000	0.05	100	90-110		
Matrix Spike (BEL0444-MS3)		**Source: 23L0700-01**			Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000	0.04	102	90-110		
Reference (BEL0444-SRM1)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0444-SRM2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		

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DELLAVALLE™
LABORATORY INC

Luis Correia Dairy
5973 Sugarplum Drive
Atwater, CA 95301

Account# 00-0021343
Account Manager: Ben Nydam
Submitted By: Luis Correia

Received: 12/13/2023 7:00
Reported: 12/20/2023 15:13

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0444 (Continued)										
Reference (BEL0444-SRM3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0444-SRM4)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		

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Received: 12/13/2023 7:00
Reported: 12/20/2023 15:13

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0614										
Blank (BEL0614-BLK1)					Prepared & Analyzed: 12/14/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	0.10		0.01	mmhos/cm						
Electrical Conductivity umhos	97.4		10.0	umhos/cm						
pH	3.2		1.0	units						
Blank (BEL0614-BLK2)					Prepared & Analyzed: 12/14/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	0.01		0.01	mmhos/cm						
Electrical Conductivity umhos	10.5		10.0	umhos/cm						
pH	4.4		1.0	units						
Blank (BEL0614-BLK3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Duplicate (BEL0614-DUP1)		**Source: 23L0743-03**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	2.12		0.01	mmhos/cm		2.12			0.165	10
pH	7.8		1.0	units		7.8			0.258	10
Electrical Conductivity umhos	2120		10.0	umhos/cm		2120			0.165	10
Duplicate (BEL0614-DUP2)		**Source: 23L0771-01**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	0.66		0.01	mmhos/cm		0.66			0.923	10
Electrical Conductivity umhos	664		10.0	umhos/cm		658			0.923	10
pH	7.5		1.0	units		7.6			0.529	10
Reference (BEL0614-SRM1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	458			umhos/cm	426.0		107	90-110		
Reference (BEL0614-SRM2)					Prepared & Analyzed: 12/14/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0614-SRM3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1150			umhos/cm	1000		115	90-110		
Electrical Conductivity umhos	1150			umhos/cm	1000		115	90-110		
Reference (BEL0614-SRM4)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEL0614-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1090			umhos/cm	1000		109	90-110		

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Reported: 12/20/2023 15:13

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0614 (Continued)										
Reference (BEL0614-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity umhos	1090			umhos/cm	1000		109	90-110		
Reference (BEL0614-SRM6)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0614-SRM7)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0614-SRM8)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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Received: 12/13/2023 7:00
Reported: 12/20/2023 15:13

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0615										
Blank (BEL0615-BLK1)					Prepared & Analyzed: 12/14/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	4.7		1.0	units						
Blank (BEL0615-BLK2)					Prepared & Analyzed: 12/14/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.6		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0615-BLK3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0615-DUP1)		**Source: 23L0774-08**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	0.53		0.01	mmhos/cm		0.53			1.05	10
pH	8.3		1.0	units		8.3			0.00	10
Electrical Conductivity umhos	528		10.0	umhos/cm		534			1.05	10
Duplicate (BEL0615-DUP2)		**Source: 23L0776-01**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	5.91		0.01	mmhos/cm		5.89			0.203	10
pH	7.0		1.0	units		7.0			0.286	10
Electrical Conductivity umhos	5910		10.0	umhos/cm		5890			0.203	10
Reference (BEL0615-SRM1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	435			umhos/cm	426.0		102	90-110		
Reference (BEL0615-SRM2)					Prepared & Analyzed: 12/14/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0615-SRM3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0615-SRM4)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0615-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		

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Atwater, CA 95301

Account# 00-0021343
Account Manager: Ben Nydam
Submitted By: Luis Correia

Received: 12/13/2023 7:00
Reported: 12/20/2023 15:13

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0615 (Continued)										
Reference (BEL0615-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0615-SRM6)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0615-SRM7)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0615-SRM8)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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12/13/23 07:00 23L0771

JG

WATER WORK REQUEST

Bill To: Acct No. 21343 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Corriea Dairy, Luis
Address: 5973 Sugarplum Drive
City, State, Zip: Atwater, CA 95301
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Luis Correia - (209) 201-1520

Ranch _____

Date sampled: 12/11/23

Sampled by: LCorreia

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

No. of Samples 3 No. Bottles 3

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES	Sampled From:	Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C	#/F
1. DW Barn	Sampled From:	12/11/23	7pm	Ø	5.4	/ -0.1
2. DW House	Sampled From:	12/11/23	710p	Ø	5.5	/ -2.3
3. IW	Sampled From:	12/11/23	725p	Ø	4.7	/ -0.1
4.	Sampled From:					
5.	Sampled From:					
6.	Sampled From:					
7.	Sampled From:					
8.	Sampled From:					
9.	Sampled From:					
10.	Sampled From:					

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

IR Thermometer SN: 221511276
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Hanford

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	LCorreia	Correia Dairy	12/11/23 725p	12/12/23 7Am
Second	mpedroso	CASinc	12/12/23 7Am	12/12/23 330p
Third	[illegible]	DL	12/12/23 3:30pm	
Fourth	[illegible]	DLI	12/13 [illegible]	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☐ Walk In ☑ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☑ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

12/13/23 07:00 23L0771

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1	1							
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT